Recitation 2 – Recitation 2 – Delays in Packet Switching

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Computer Networks

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Delays in Packet Switching: Four Types

- **Processing Delay**
  - Time required to examine the packet's header, checksum, etc.
- **Queuing Delay**
  - Time packet waits in the output buffer
- **Transmission Delay:** L/R
  - L = packet length (bits)
  - R = link bandwidth (bps)
- **Propagation Delay:** d/s
  - d = length of physical link
  - s = propagation speed in the medium

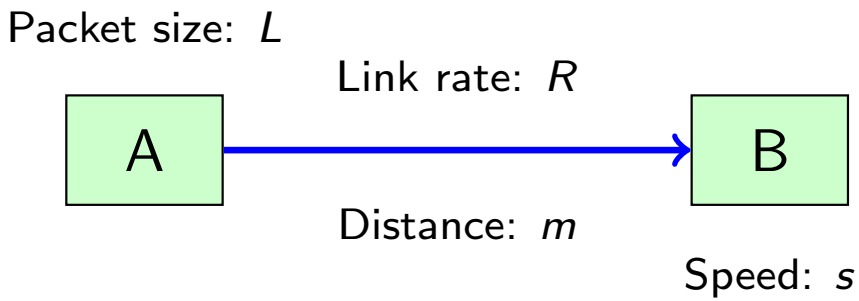
Problem 1A: Equating Transmission & Propagation Delay

Consider two hosts, A and B, connected by a single link that supports a transmission rate of R bps.

Suppose that the two hosts are separated by m meters, and the propagation speed along the link is s m/sec.

Host A sends a single packet of size L bits to host B.

Question: What is the transmission rate R for which the propagation delay equals the transmission delay?



Problem 1A: Solution

Setting propagation delay equal to transmission delay:

Propagation delay = Transmission delay

Problem 1A: Solution

Setting propagation delay equal to transmission delay:

Propagation delay $=$ Transmission delay

$$\frac{m}{s} = \frac{L}{R}$$

Problem 1A: Solution

Setting propagation delay equal to transmission delay:

Propagation delay $=$ Transmission delay

$$\frac{m}{s} = \frac{L}{R}$$

Solving for R:

Answer

$$\boxed{R = \frac{Ls}{m}}$$

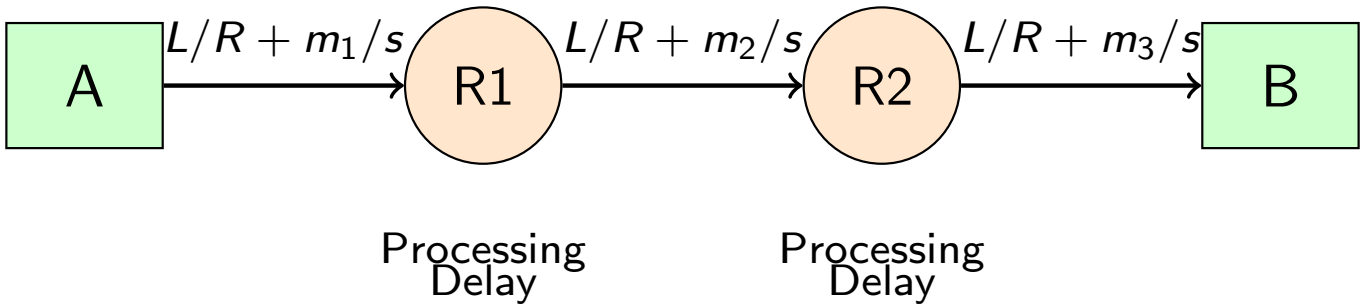
Problem 1B: End-to-End Delay Calculation

Consider two hosts, A and B, separated by three links. These three links are connected by two routers. Each router introduces a constant processing delay for each packet.

Consider sending a packet of length **1500 bytes** from A to B. Assume the propagation speed on all three links is 2.5×10^8 m/sec, the transmission rates of all three links are **3 Mbps**, the router processing delay is **3 msec**, the length of the first link is **5000 km**, the length of the second link is **4000 km**, and the length of the last link is **1000 km**.

Question: For these values, what is the end-to-end delay?

Problem 1B: Network Diagram & Given Values



Given Values

- $L = 1500$ bytes $= 12{,}000$ bits
- $R = 3$ Mbps $= 3{,}000{,}000$ bits/sec
- $s = 2.5 \times 10^8$ m/sec
- $m_1 = 5{,}000{,}000$ m, $m_2 = 4{,}000{,}000$ m, $m_3 = 1{,}000{,}000$ m
- processing delay $= 3$ msec $= 0.003$ sec

Problem 1B: Solution (Detailed Method)

Calculate each component:

$$L/R = \frac{12{,}000}{3{,}000{,}000} \text{ sec} = \frac{4}{1{,}000} \text{ sec} = 4 \text{ msec}$$

$$m_1/s = \frac{5{,}000{,}000}{2.5 \times 10^8} = 20 \text{ ms}$$

$$m_2/s = \frac{4{,}000{,}000}{2.5 \times 10^8} = 16 \text{ ms}$$

$$m_3/s = \frac{1{,}000{,}000}{2.5 \times 10^8} = 4 \text{ ms}$$

$$\text{processing delay} = 3 \text{ ms}$$

Total end-to-end delay:

$$L/R + m_1/s + \text{proc} + L/R + m_2/s + \text{proc} + L/R + m_3/s$$

$$= 4 + 20 + 3 + 4 + 16 + 3 + 4 + 4$$

Problem 1B: Solution (Detailed Method)

Answer

$= \boxed{58 \text{ ms}}$

Problem 1B: Solution (Optimized Method)

Optimized calculation (using symmetry):

$L/R = 4$ msec [Note 1: L/R is the same for all three links]

$\Sigma m = 10{,}000{,}000$ meter [Note 2: s is the same for all three links]

processing delay $= 3$ ms [Note 3: same for both routers]

Total:

$3 \times L/R + \Sigma m/s + 2 \times$ processing delay

$= 3 \times 4 + 40 + 2 \times 3$

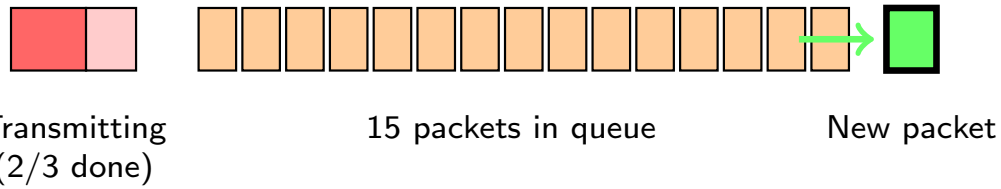
$= 12 + 40 + 6$

Answer

$= \boxed{58 \text{ ms}}$

Problem 1C: Queuing Delay with Partial Transmission

A router receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet has completed **2/3 of its transmission** on this outbound link and **15 other packets are waiting** to be transmitted (in a queue). Packets are transmitted in order of arrival. Suppose all packets are **1300 bytes** and the transmission rate is **2 Mbps**.



Question: What is the queuing delay for the new packet?

Problem 1C: Solution

Analysis:

We have 15 packets in the queue, and an additional $\frac{1}{3}$ of a packet to transmit (the current packet is $\frac{2}{3}$ done).

Total packets worth of time: $15\frac{1}{3}$ packets

Problem 1C: Solution

Analysis:

We have 15 packets in the queue, and an additional $\frac{1}{3}$ of a packet to transmit (the current packet is $\frac{2}{3}$ done).

Total packets worth of time: $15\frac{1}{3}$ packets

Calculation:

Time to transmit one packet: $\dfrac{1{,}300 \times 8}{2{,}000{,}000}$ sec

Problem 1C: Solution

Analysis:

We have 15 packets in the queue, and an additional $\frac{1}{3}$ of a packet to transmit (the current packet is $\frac{2}{3}$ done).

Total packets worth of time: $15\frac{1}{3}$ packets

Calculation:

Time to transmit one packet: $\dfrac{1{,}300 \times 8}{2{,}000{,}000}$ sec

Total queuing delay: $\dfrac{1{,}300 \times 8 \times 15\frac{1}{3}}{2{,}000{,}000}$ sec

Answer

$= \boxed{79.733 \text{ ms}}$

Packet Sniffing: Concepts & Tools

- **Intercept traffic going over the network (libpcap)**
  - Analyze protocols
  - Reconstruct segments, messages and entire connections
  - **Promiscuous mode** (wired): capture all packets on network segment
  - **Monitor mode** (Wi-Fi): capture raw 802.11 frames
  - Both require admin privileges (sudo or capabilities)
- **Common packet sniffing tools**
  - **tcpdump**: Command-line packet analyzer (we'll use this!)
  - **Wireshark**: GUI-based analyzer (homework assignment)
- **PCAP file format**
  - Stores recorded network traffic
  - Can be analyzed later (forensics, troubleshooting, learning)

tcpdump Demo: Basic Commands

Start capturing packets (filter specific hosts):

```
sudo tcpdump -i any 'host httpbin.org or host ynet.co.il' \
  -w capture.pcap
```

In another terminal, generate HTTP traffic:

```
curl -v http://httpbin.org/html
```

Then generate HTTPS traffic:

```
curl -v https://ynet.co.il
```

Stop capture (Ctrl+C) and read the file:

```
tcpdump -r capture.pcap -A
```

tcpdump Demo: HTTP vs HTTPS Comparison

What we see in HTTP (httpbin.org/html):

- HTTP GET request in plaintext
- HTTP headers (Host, User-Agent, etc.)
- Full HTML response content - completely readable!

What we see in HTTPS (ynet.co.il):

- TLS handshake (Client Hello, Server Hello)
- **HTTP content: ENCRYPTED** - unreadable!

Security takeaway: HTTP is plaintext (anyone can read it), HTTPS encrypts all content. Always use HTTPS for sensitive data!

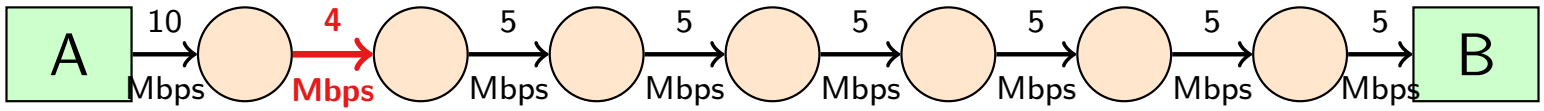
Advanced Problem: Multi-Packet File Transfer

We wish to send a file of size **100 Mbit** from source A to destination B. There are exactly **7 routers** on the only existing route between A and B. All links along the route support a transmission rate of **5 Mbps**, except the first link which supports **10 Mbps** and the second link which supports **4 Mbps**.

The propagation speed is 4×10^8 m/sec and the total distance is 10^8 m. The file is divided into **5 equal-size packets** for transmission (each of size **20 Mbit**).

Question: How much time in seconds will it take for the last bit of the file to reach host B?

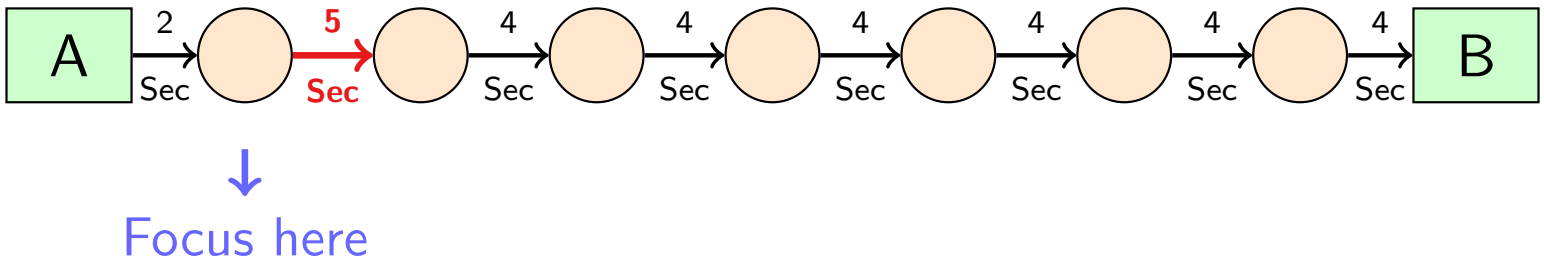
Step 1: Understanding the Network Topology



Note: The 4 Mbps link is the bottleneck!

Question: What is the Transmission Delay for a packet (20 Mbit) on each link?

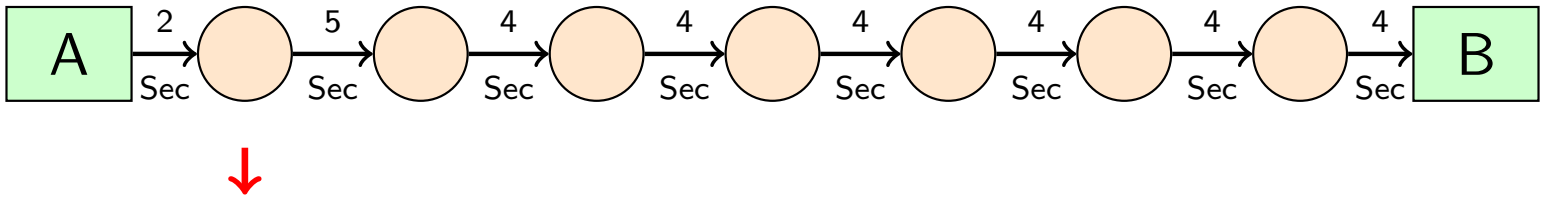
Step 2: Calculate Transmission Delays per Link



When will the fifth & last packet reach the first router?

Where will the other packets be at that time?

Step 3: Analyze Packet Timeline at Router 1

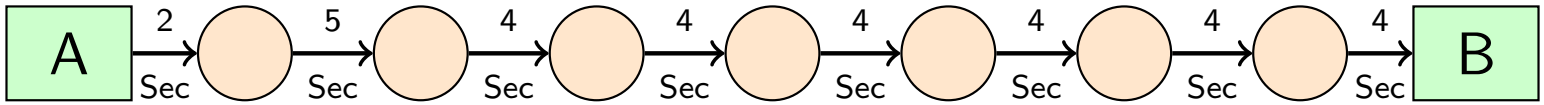


Time (s)	A	R1	R2	R3	R4	R5	R6	R7	B
0	P1-5								
10		<P2>, P3, P4, **P5**	P1						

<P2> is being transmitted (3/5 done)
P3, P4 are in queue
P5 just arrived

What is the Queuing Delay for P5 on the first router?

Step 4: Calculate Queuing Delay for P5



The Queuing Delay for P5 on the first router is:

Wait for P2 to finish: $\frac{2}{5} \times 5 = 2$ sec

Wait for P3: 5 sec

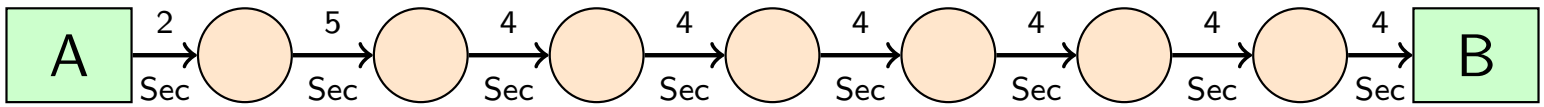
Wait for P4: 5 sec

Queuing Delay for P5

$\left(\frac{2}{5} + 2\right) \times 5 = \boxed{12 \text{ sec}}$

Will P5 incur any additional Queuing Delays along its remaining route? Why?

Step 5: Why No Additional Queuing Delays?

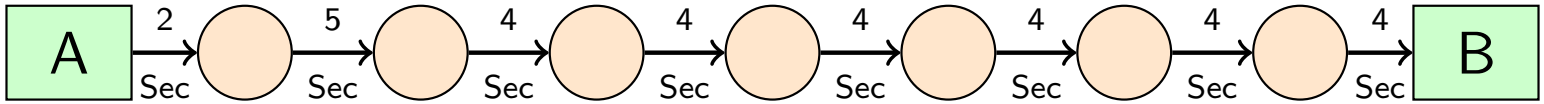


Will P5 incur any additional Queuing Delays along its remaining route?

Important Insight

No. The transmission rate along the remaining route is always $\leq$ 5 Mbps (not slower than the bottleneck). Packets will spread out further apart, so no additional queuing occurs.

Step 6: Final Calculation



How much time in seconds will it take for the last bit of the file to reach host B?

Propagation Delay: $\frac{10^8}{4 \times 10^8} = 0.25$ sec

Total:

Send all packets to R1: 10 sec

Queuing at R1: 12 sec

Transmit on bottleneck: 5 sec

Transmit on remaining 6 links: $4 \times 6 = 24$ sec

Propagate: 0.25 sec

Step 6: Final Calculation

Final Answer

$$10 + 12 + 5 + 24 + 0.25 = \boxed{51.25 \text{ sec}}$$

Key Takeaways

- **Four Types of Delays:**
  - Processing, Queuing, Transmission (L/R), Propagation (d/s)
- **Key Concepts:**
  - End-to-end delay = sum of all component delays
  - Bottleneck links determine maximum throughput
  - Queuing occurs when arrival rate $>$ service rate
- **Tools:**
  - tcpdump for packet analysis
  - Understanding protocol layers (frame, datagram, segment, message)
- **Problem-Solving Strategy:**
  - Identify bottleneck
  - Calculate delays component by component
  - Track packet progression through the network